Investigating Social Media Posts Across Mental Health Categories

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STATS 133: Introduction to Text Mining

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I. Abstract

This study investigates whether linguistic and sentiment-based features from social media posts can be used to classify mental health categories. By using a dataset of over 53,000 posts labeled with conditions such as depression, anxiety, and suicidal status, we were able to process text data with TF-IDF, bigrams, sentiment lexicons, AFINN, Bing, NRC, and topic modeling. We chose to leverage two supervised learning models, multinomial logistic regression and random forest, to predict mental health labels based on the outlined features. Results indicate that the random forest model significantly outperformed logistic regression, achieving an accuracy of 78.1% compared to 52.7%, and also demonstrated stronger performance across all classes. These results suggest that combining sentiment, linguistic, and topic-based features can effectively support the classification of mental health signals in social media, with potential applications in real-time mental health monitoring and early intervention.

II. Introduction

The 21st century has seen the rise of social media platforms that dominate online spaces, with platforms such as Twitter, Reddit, and Instagram, and many others. Social media allows users on these platforms to comment, post, or discuss with others on a wide array of topics, ranging from casual and humorous to serious and political. Users of online spaces can form communities or discuss their emotions and feelings on the internet. Simultaneously, as discussions regarding mental health continue to evolve in society, social media provides a glimpse into people's minds. Rather than directly surveying people on their mental health status, social media posts provide a log of people's emotional status. Social media usage itself can also affect people's psychological and mental well-being. We can study these posts to analyze people's mental health over time. The purpose of this study is to use sentiment and linguistic analysis to investigate if we can classify certain mental health conditions by using various machine learning techniques. Our main research question is "Can linguistic and sentiment features extracted from social media posts be used to distinguish and predict different mental health categories by their associated text?"

III. Methods and Procedures

3.1 Participants

Our study's unit of analysis are individual social media posts from the publicly available Sentiment Analysis for Mental Health dataset on Kaggle. This dataset stores 53,043 anonymous social media posts collected from various sources such as Reddit, X, and additional social media platforms. Each entry is labeled with mental health-related categories such as "depression," "anxiety," "stress," "suicidal," "personality disorder," "normal," and "bipolar." It includes a "normal" label that is associated with the posts with no negative mental health indicators. Demographic information, such as age and gender, is not available in this dataset. The inclusion criteria are posts written in English with complete text content, whereas posts with missing text, duplicates, or fewer than five words will be excluded. The distribution of posts can also be seen in figure 2, which displays that some classes are represented more in the dataset than others. As a result, the dataset reflects a diverse range of linguistic expressions and emotional content across different mental health conditions. However, the lack of demographic information limits the ability to examine how these patterns may vary across specific populations.

unique_id	statement	status
1	oh my gosh	Anxiety
2	trouble sleeping, confused mind, restless heart. All out of tune	Anxiety
3	All wrong, back off dear, forward doubt. Stay in a restless and restless place	Anxiety
4	I've shifted my focus to something else but I'm still worried	Anxiety
5	I'm restless and restless, it's been a month now, boy. What do you mean?	Anxiety
6	every break, you must be nervous, like something is wrong, but what the heck	Anxiety
7	I feel scared, anxious, what can I do? And may my family or us be protected :)	Anxiety
8	Have you ever felt nervous but didn't know why?	Anxiety
9	I haven't slept well for 2 days, it's like I'm restless. why huh :([.	Anxiety
10	I'm really worried, I want to cry.	Anxiety

Figure 1: Glimpse of Dataset (53,043 Observations x 3 Variables)

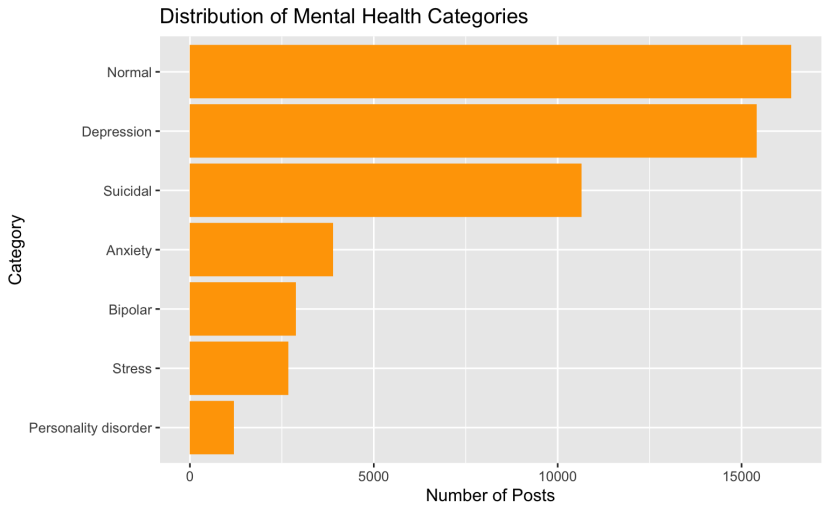

Figure 2: Overview of Distribution of Posts Across Mental Health Categories

3.2 Design

This study uses a quantitative correlational design. The purpose is to investigate the associations between sentiment patterns, text elements, and mental health categories, as provided by the social media posts in the dataset. Sentiment will be measured both as a continuous polarity score and as categorical classifications utilizing a range of lexicons. The primary target variable is the mental health status which we will be attempting to classify. Independent variables will include structural features such as word count, unigram and bigram TF-IDF features, sentiment data from AFINN, Bing, and NRC lexicons, and topic modeling using LDA. The sentiment scores are computed at the post level and compared across mental health categories.

We selected this data set as it provided a comprehensive selection of posts across various social media platforms. These posts include a wide range of expressions and emotions across textual data, which allow deeper insight into understanding and predicting mental health conditions. As such, our study aims to analyze mental health trends for academic research, raising awareness for individuals in crisis, and developing mental health support.

3.3 Materials and Procedures

The dataset was downloaded as a CSV file from Kaggle and imported into R. The dataset was inspected, and we used data cleaning to remove incomplete entries, duplicates, and posts not meeting the inclusion criteria. Following data cleaning, we initiated text preprocessing by converting all the text to lowercase, removing punctuation and stop words, normalizing apostrophes, removing urls and usertags(@), removing hashtags, keeping spaces/letters/apostrophes, and tokenizing sentences into individual words, while keeping alphabetic tokens. The tokens were joined with the AFINN, Bing, and NRC lexicons, and sentiment scores were separated at the post level. We also chose to use a log transformation on the AFINN sentiment scores to ensure reliable analysis. Then, descriptive statistics and visualizations were generated using ggplot and wordclouds, to compare sentiment and keywords across mental health categories. In addition to sentiments, term frequency (TF) and term frequency-inverse document frequency (TF-IDF) were measured across all text data. Topic modeling via LDA, Latent Dirichlet Allocation, and bigram features were collected and combined with additional sentiment features into a single features data frame to be used for our predictive model. Finally, the deployment of Random Forest from the "randomForest" package and multinomial logistic regression model from "nnet" were used to provide predictions for mental health class labels, given sentiment scores and additional metadata.

3.4 Instrument

All the pre-model analysis and data wrangling was done with R using the tidyverse and tidytext packages. The text was tokenized using 'unnest_tokens()' from the tidytext package. The sentiment was measured using three lexicons which are AFINN, Bing, and NRC. AFINN, which falls on a scale -5/ +5 with larger negative scores being extremely negative sentiments, and more positive scores being positive sentiments, was used to assign continuous polarity scores. Bing was used to classify words as positive or negative, and NRC helped categorize words into emotional dimensions such as anger, fear, and joy. These three lexicons are widely used in text analysis and accessible through tidytext. Harnessing various lexicons improves the reliability of the investigation and allows a comprehensive comparison across mental health labels.

To model mental health classification, two supervised learning approaches were implemented: multinomial logistic regression and random forest. The dataset was split into training (80%) and testing (20%) sets using stratified sampling to preserve class distributions. Multinomial logistic regression was fit to model relationships between features and categorical mental health labels. A random forest model with 300 trees to capture nonlinear relationships and interactions among features. Model inputs included structural features like word count, sentiment features including AFINN, Bing, NRC, and log-transformed AFINN, unigram and bigram TF-IDF representations, and topic probabilities derived from Latent Dirichlet Allocation. These were all compiled into a DTM in order to apply the data to the logistic regression model and Random Forest. Model performance was evaluated on the test set using accuracy, confusion matrices, sensitivity, specificity, and Cohen's kappa, allowing comparison between approaches and assessment of classification effectiveness across mental health categories.

IV. Results

4.1 Interpreting the Data

Initial exploration of the dataset reveals clear differences in linguistic and sentiment patterns across mental health categories. TF-IDF analysis shows that depression, anxiety, and suicidal posts contain more emotionally intense and negative language, while "normal" posts are more neutral. Sentiment results from AFINN and Bing further support this, with these categories exhibiting more negative polarity scores, whereas normal posts cluster closer to neutral values. NRC emotion analysis also indicates higher levels of sadness, fear, and anger in depression and anxiety posts compared to lower emotional intensity in normal posts. Overall, these patterns suggest that textual and sentiment-based features capture meaningful differences across categories, supporting their use in classification models. Additionally, variability in sentiment distributions across categories suggests that certain conditions, such as bipolar disorder, exhibit a wider range of emotional expression compared to more consistently negative categories like depression. The observed overlap between some categories, particularly depression and suicidal labels, shows that while differences exist, linguistic indicators may not always be very separable. This highlights the inherent complexity of mental health classification

based solely on textual data. Despite this, the consistent directional trends across multiple sentiment measures reinforce the strength of these features. Together, these findings provide a strong foundation for applying machine learning models to assess whether these patterns can be leveraged for accurate classification.

4.2 Exploratory Data Analysis

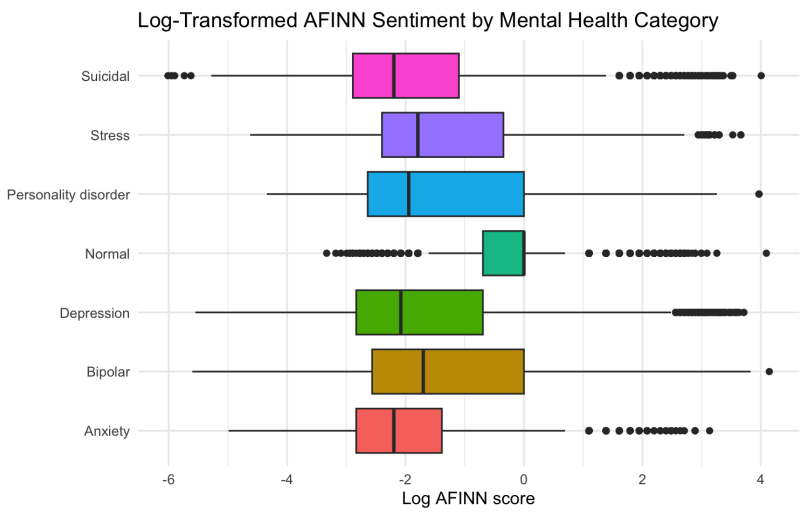

Figure 3: Log Transformed AFINN Sentiment Across Mental Health Classes

Looking at the log transformed Afinn scores we gain insights on the overall sentiments across each category. Most categories mostly displayed a negative sentiment to neutral scores as the Afinn scores range from -6 + 4, which are slightly out of bounds from the standard Afinn range of -5/ +5 due to the effects of the log transformation. Suicidal, Depression, and Anxiety had the lowest median scores meaning that they display the strongest negative sentimentins across their text entries. This is an unsurprising result because these mental health conditions are typically associated with negative self-worth, mood swings, and other difficult sentiments. Personality disorder and Stress displayed moderate to low negative sentiment levels, suggesting that these categories have less negative sentiments compared to other categories such as Bipolar status had largest amount of variability in terms of sentiment scores, which is also to be expected, given bipolar behaviors typically vary

from extreme high and extreme low mood conditions. The normal category had the most neutral sentiment, as its median was closest to 0, indicating that its language was fair and not biased towards positive/negative sentiments. However, across all groups there are positive outliers present, which indicates that positive sentiments are present regardless of the mental health status, but the majority of the categories typically associated with negative sentiments.

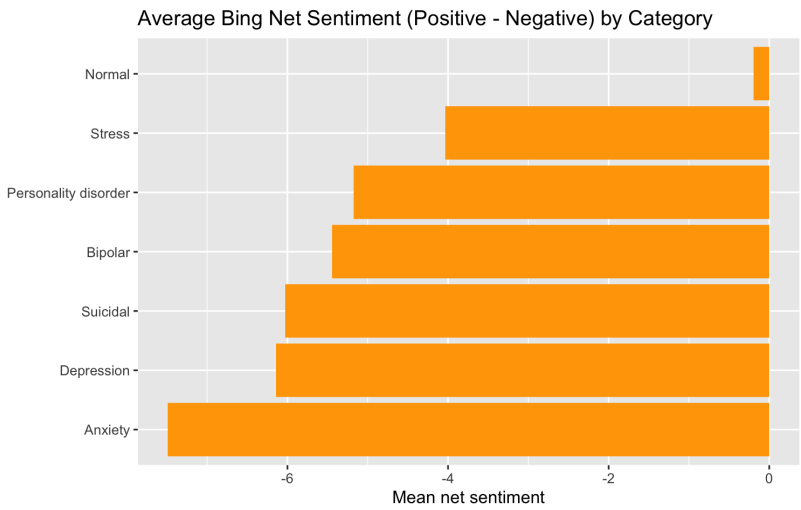

Figure 4: Average Bing Net Sentiments Across Mental Health Classes

The average net bing net sentiments reveal that negative sentiments are most prevalent across mental health categories. Again, we notice Anxiety, Depression, and Sucidal categories lead in overall negative sentiments indicating that they contain the most negative expression, matching our previous Afinn results. Personality disorder and Bipolar follow indicating that they do have moderate amounts of negative sentiments, however there are more positive words compared to the Anxiety, Depression, and Suicidal. Stress had moderate to low levels of negative sentiments, indicating that there is less of a presence of strong negative terms compared to the other categories. Normal had the lowest levels of negative sentiments, with normal having the least negative sentiments by far, indicating that it is balanced between both positive and negative terms. These patterns suggest that sentiment polarity is a strong distinguishing feature between classes, with more severe conditions exhibiting more negative language. The differences observed across

categories, from highly negative, depression and suicidal, to more neutral, normal, indicates that sentiment may capture differences in emotional intensity rather than strictly discrete classes. Furthermore, the presence of both positive and negative terms across all categories highlights the nuance of mental health expression, where individuals may express mixed emotions within a single post. This reinforces the idea that while sentiment is informative, it should be combined with other textual features to improve classification performance.

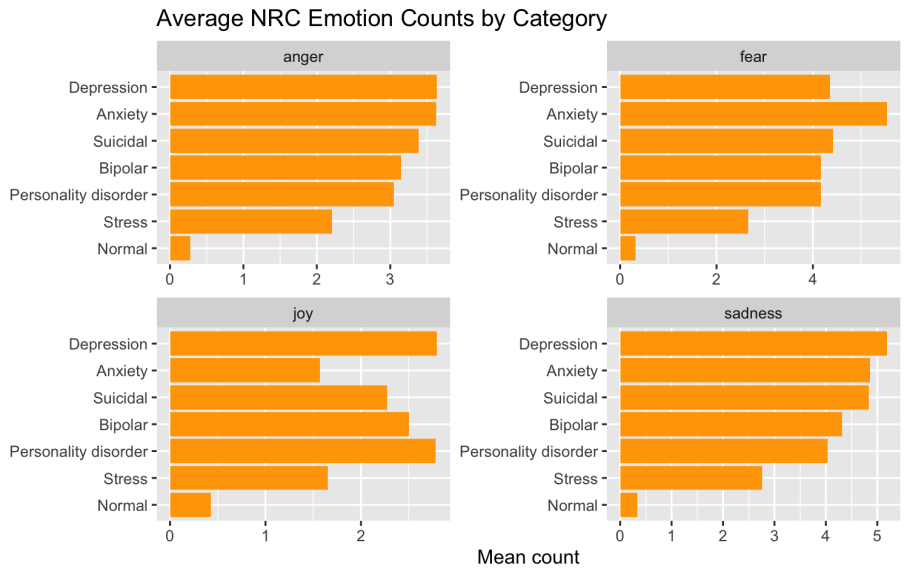

Figure 5: Average NRC Sentiments Across Mental Health Categories

Using the NRC lexicon, we gain insights on which categories have the highest associations of words between emotions such as anger, fear, anticipation, trust, and joy. Across the sentiments of fear, anger, and sadness, depression, anxiety, and suicidal subgroups consistently rank among the top three counts. This can be taken to mean these three mental health conditions likely experience these negative emotions most strongly. Particularly, for both anger and sadness, depression scored the highest count, while anxiety scored the highest in fear and roughly tied with depression in anger. This shows that while anger and sadness are predominant emotions in depressive status, anxiety status tend to struggle with fear and anger. Interestingly, in the joy sentiment, depression status scored the highest mean count, followed

by personality disorder, bipolar, then suicidal statuses. Though this differs from our understanding of the mental health conditions, this count may come from these groups recalling happy memories or things they enjoy, as a coping mechanism away from trauma or struggles.

Conversely, for normal status, they scored the lowest mean count across the board, in joy, anger, sadness, and fear categories. This illustrates that normal status does not experience emotions on as intense a level as users with mental health conditions. Additionally, normal status scored the highest on the joy sentiment, and scored similarly for other negative emotions. This shows that normal people are more likely to post and engage in a positive manner, and less likely to express negative emotions. Stress status, additionally, scored second-lowest on all the sentiments, though the mean counts compared to normal status was significantly higher. While stress status points to a weakened mental wellbeing, compared to normal status, stress status mean counts still remain significantly lower than actual mental health conditions. As such, stress status may cause heightened emotions, but other mental health conditions still reveal stronger emotional expressions.

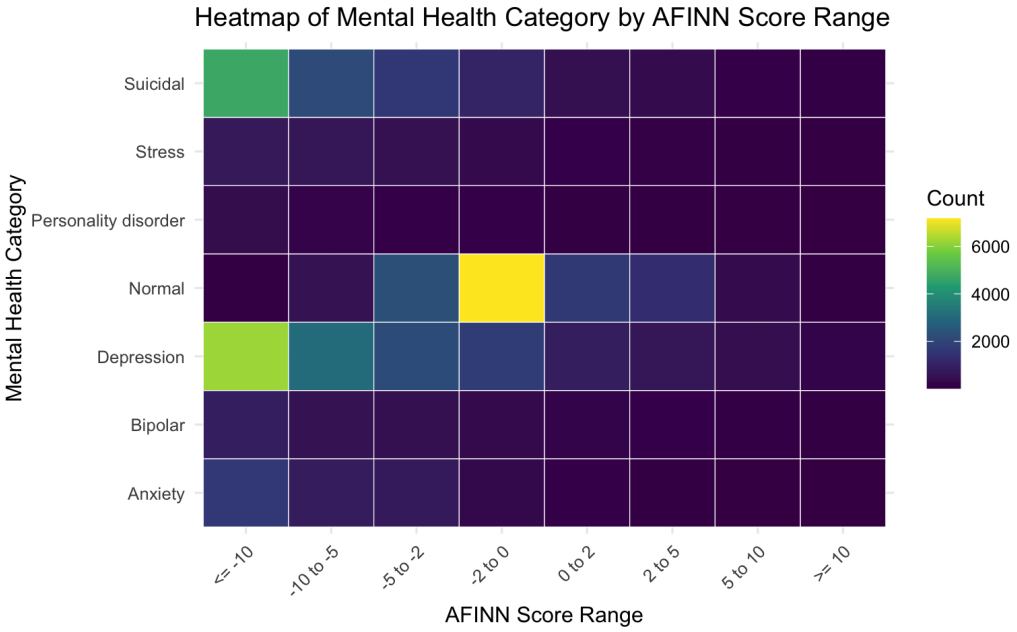

Figure 6: Heatmap of Mental Health Categories by Afinn Score Range

The heatmap reveals that most posts across mental health categories fall in ranges of less than -10 to -5 to 2. When viewing each of the categories of the heatmap, Depression appears to have the largest count and highest concentration of posts with negative sentiments. Following Depression, Anxiety and Suicidal follow also having significantly large counts of negatively expressed posts, though not as extreme as compared to Depression. These three categories again reflect the similar pattern as seen from the previous analysis of other lexicons, where Depression, Suicidal , and Anxiety lead overall in terms of expressing the strongest counts of negative sentiments. Stress, Personality disorder, and Bipolar statuses had lower counts, but their sentiments generally ranged in moderate levels of negative sentiments, -5 to -2, but not as extreme compared to the previous categories. Normal category ranged from -2 to 0, which again displayed neutral sentiment and having mix between positive and negative expressed words, ultimately leading in overall highest count of positive sentiment words.

4.3 Modeling Results

The multinomial logistic regression model achieved an overall accuracy of 52.7% with a kappa of 0.349, which indicates a moderate agreement beyond chance. The model performed best in predicting the "Normal" class, with a high sensitivity score of 0.933, and showed decent performance for "Depression" with a sensitivity score of 0.662. However, performance was notably poor for several minority classes, including "Bipolar," "Stress," and "Personality disorder," where sensitivities were close to zero. The confusion matrix further reveals substantial mis-classification, particularly between "Depression" and "Suicidal" categories, suggesting difficulty distinguishing between closely related mental health conditions. Overall, the logistic regression model struggled to capture the complexity of the data, particularly for less frequent classes and more nuanced distinctions between categories.

The random forest model demonstrated a much better performance, achieving an accuracy of 78.1% and a kappa of 0.713, which emphasizes a strong agreement beyond chance. The model performed well across most classes, with high sensitivity for "Depression" with 0.835, "Normal" with 0.926, and "Anxiety" with a score of 0.775, as well as strong performance for "Suicidal" with a score of 0.666. While logistic regression struggled with minority classes, the random forest was able to effectively capture minority classes such as "Bipolar" with 0.731 and "Personality disorder" scoring 0.639. The

confusion matrix shows fewer misclassifications, but some overlap remains between "Depression" and "Suicidal," reflecting their linguistic similarity and difficulty to classify. These results suggest that the random forest model better captures nonlinear relationships and interactions among features. This led to more accurate and balanced classification across mental health categories.

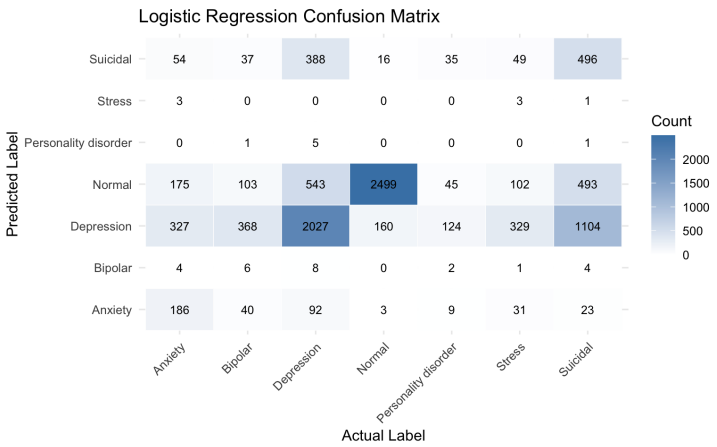

Figure 7: Logistic Regression Confusion Matrix

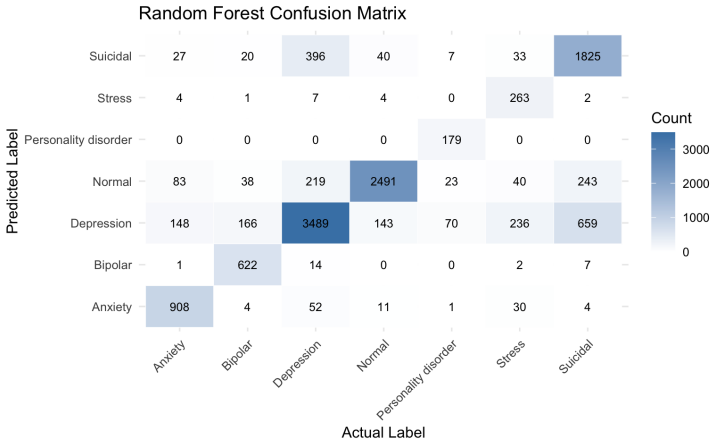

Figure 8: Random Forest Confusion Matrix

V. Discussion

5.1 Interpretation of Findings

The findings of this study show that text and sentiment based features extracted from social media posts provide meaningful signals for classifying between mental health categories. Consistent patterns observed across TF-IDF, sentiment analysis, and emotion lexicons show that categories such as depression, anxiety, and suicidal have more negative and emotionally intense language, while normal posts tend to remain more neutral. These distinctions suggest that emotional expression and language use are in fact associated with mental health status.

The modeling results further emphasize this relationship, as the random forest model achieved substantially higher accuracy than multinomial logistic regression. This discrepency suggests that the relationships between features and mental health categories are complex and nonlinear, which means models capable of capturing interactions between linguistic, sentiment, and thematic features are necessary. The inclusion of bigrams and topic modeling also contributed to improved performance, indicating that contextual and phrase-level data has an important role beyond individual word frequencies.

However, misclassification between closely related categories, particularly depression and suicidal, highlight the overlap in how these conditions are expressed in text. This suggests that mental health categories may exist along a spectrum rather than as clearly separable groups. Overall, the results demonstrate that combining multiple feature types provides a strong foundation for classifying mental health categories from social media posts.

5.2 Limitations and Improvements For Future

Despite our promising results, there are still limitations that should be addressed. First of all, the dataset lacks demographic information such as age, gender, and background, which limits the ability to understand how mental health conditions may vary across different key populations. Secondly, social media text is inherently noisy and informal, which probably introduces inconsistent language which can reduce the models accuracy. Also, the labels in the dataset may not perfectly reflect true mental health conditions, as they are based on text rather than clinical diagnosis. Another limitation is the reliance on lexicon-based sentiment methods, which may fail to capture sar-

casm, context, or deeper semantic meaning. Even though bigrams and topic modeling help address some of these issues, more advanced methods such as contextual embeddings, like BERT, could further improve performance. lastly, class imbalance within the dataset could potentially influence model performance, especially for less frequent categories.

Future improvements could focus on incorporating richer contextual models, balancing class distributions, and harnessing additional metadata to improve classification performance. Also, expanding the investigation to include temporal elements or longitudinal data could also provide deeper insights into how mental health expressions evolve over time.

VI. Conclusion

In conclusion, our study found strong signals and interactions between sentiment-based features extracted from social media posts, and mental health categories. As social networks continue to grow as a tool for self-expression and exchange of ideas, we hope the results will allow for better awareness of online crisis signals. Specifically, we hope the data can help train social media AI or algorithms to better identify posts that require flagging, banning, or even intervention. We expect social media companies, using similar models and making appropriate adjustments, alongside combining additional demographic and personal information, can better address posts key issues affecting those across an age or gender spectrum.

The overwhelming amounts of social media posts that indicate various mental health categories also demonstrate a worrisome trend. Although it is great for people to articulate their feelings, venting online is not a complete substitute for other forms of treatment or help. In some cases, speaking in echo chambers online may even exacerbate one's condition, as online users are not as equipped as psychiatric professionals, or reinforce a user's negative thought patterns. What's worse, consuming social media content continuously can lead to burnout and a further decline in one's mental health conditions. The fear is that people with mental health conditions will depend solely on social media, hiding behind an anonymous identity, and isolate themselves from seeking help in the real world. Therefore, we hope the results of the paper can aid those in need, and alert families of users showing concerning online behavior, to reach out and check in on their mental health.

VII. Resources

1. Almohalwas, A (2026). *Modules.* BruinLearn at University of California, Los Angeles. https://bruinlearn.ucla.edu/courses/225097/modules

2. Sarkar, Suchintika. "Sentiment Analysis for Mental Health." Kaggle, 5 July 2024, www.kaggle.com/datasets/suchintikasarkar/sentiment analysis-for-mental-health.